Joy Chang

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Education

University of Washington, Seattle **Jun 2028**

B.S. in Informatics, Data Science Concentration; Minor in Statistics *GPA: 3.91/4.0*

Relevant Courses: Data Structures & Algorithms, Machine Learning, Database Management, Client-Side Web Development

Leadership: Women in Informatics – Diversity Committee Lead, Husky Coding – Lead Front-End Engineer

Experience

Lynden Inc. **Jun 2026 – Sep 2026**

DevOps Engineer Intern *SeaTac, WA*

- Engineered a Python-based repository migration workflow using the GitHub REST API and PyGithub to automate the transfer of 800+ repositories, reducing manual migration effort by 80%.
- Developed Python scripts to automate system health checks across 5+ servers and networked devices, streamlining infrastructure monitoring and reducing repetitive diagnostic work by 7 hours per week.
- Diagnosed and resolved TCP/IP, DNS, and DHCP connectivity issues across enterprise systems, restoring service for affected users and documenting resolutions to improve future troubleshooting efficiency.

User Empowerment Lab, UW Information School **Mar 2025 – Present**

Machine Learning Research Assistant *Seattle, WA*

- Conduct AI/ML and HCI research examining real-world LLM behavior, user interaction patterns, and AI-assisted decision-making.
- Developed an LLM-powered chatbot using OpenAI GPT and Python/FastAPI to evaluate safer responses to sensitive teen–AI conversations.
- Clean and analyze conversation datasets using Python, Pandas, and feature engineering to identify usage trends.

Sensoria Health Inc. **Sep 2023 – Jun 2024**

Software Development Intern *Redmond, WA*

- Built scalable data pipelines integrating smart-textile wheelchair sensors with an iOS application, supporting 100+ devices.
- Developed Swift-based interfaces for real-time wheelchair pressure monitoring and hourly repositioning alerts, reducing caregiver intervention time by 40%.
- Optimized sensor power management through sleep-mode functionality, extending battery life by 30% while maintaining secure data communication; contributed to Agile development, code reviews, and CI/CD.

Projects & Publications

Husky Maps | *Java, Graph Algorithms, Data Structures*

- Developed a Java-based navigation and search system, implementing BFS, DFS, and Dijkstra's algorithms for graph traversal and shortest-path route finding.
- Built autocomplete using BST, TST, and LLRB data structures, processing 1K+ queries under 50ms, 5x faster than baseline.
- Engineered priority queue variants to optimize graph operations and reduce pathfinding latency by 30%.

WCAG Violation Classification Model | *Python, scikit-learn, NLP*

- Built a multi-class NLP classification pipeline to map HTML accessibility violation descriptions to corresponding WCAG guideline categories using Logistic Regression.
- Preprocessed accessibility data and engineered 10K TF-IDF features with n-grams, achieving 89% macro F1-score across 15 WCAG categories on a held-out test set.

Wordplay – Open Source | *TypeScript, Svelte, GitHub*

- Contributed to Wordplay, an open-source programming language and web platform designed to make programming more accessible and inclusive.
- Developed a multilingual feedback interface using TypeScript and Svelte, enabling on-demand translation of user-submitted reports to improve accessibility for non-English-speaking users.

How Are Emerging Adults Using ChatGPT in the Wild? | *AI/ML Research, NLP* **AIES 2026 – Accepted**

- Co-authored research accepted to AIES 2026, analyzing 157K+ ChatGPT messages from 85 emerging adults across 17 countries to investigate real-world AI usage and social support patterns.
- Analyzed longitudinal conversation data using descriptive statistics and LLM-assisted classification, contributing to the identification of 10 user prompt categories and 5 AI response patterns.

Technical Skills

Languages: Python, Java, SQL, Swift, R, JavaScript, TypeScript, HTML/CSS

AI/ML & Data: Pandas, NumPy, scikit-learn, PyTorch, Matplotlib, Natural Language Processing, Statistical Analysis, Tableau

Developer Tools & Frameworks: Git, GitHub, FastAPI, Azure DevOps, Docker, React, Svelte, JUnit, Jira, VS Code, Jupyter, RStudio, Xcode, CI/CD